

5(a)	<i>ovale / malariae</i> ; A knowlesi must be spelled correctly <i>if both given, must be correct spelling</i>	1
5(b)(i)	any two from: <i>details</i> ; pH approx. 7.4 <u>and</u> 37 °C A pH7 / neutral pH A 36 / 38 °C pH of, plasma / blood or body temperature or use conditions occurring in, body / plasma / blood / humans ; <i>idea that</i> oxygen uptake / haemoglobin activity / AW, affected by, temperature / pH ;	2
5(b)(ii)	any three from: <i>in a person with malaria</i> 1 <u>affinity</u> of haemoglobin for oxygen, decreases / AW ; <i>context is infection so R in presence of carbon dioxide</i> 2 lower (percentage) saturation of haemoglobin with oxygen ; <i>allow ecf from mp1</i> A less oxygen binds to haemoglobin (in the lungs) A haemoglobin loads less oxygen AW 3 <i>idea that</i> <u>more difficult</u> to load oxygen in the lungs ; I slower 4 (as) higher partial pressure of oxygen needed to reach same percentage saturation of, haemoglobin ; 5 <i>idea that</i> oxygen released more readily in respiring tissues or in tissues, more oxygen released at any one partial pressure ; I carbon dioxide but R if stated that carbon dioxide causes oxygen to be released more readily	3

Question	Answer	Marks
5(c)	any four from: (diameter / width) 6–8 μm so able to pass through (lumen of) capillaries ; I small size - small size / small diameter / just fits in a capillary / AW, so in a line / one by one / AW, which slows blood flow / maximises time to take up oxygen / reduces diffusion distance for uptake of oxygen ; A gas exchange for oxygen - no, nucleus / mitochondria ; A no organelles - so more space for haemoglobin ; AW <i>must be linked to mp3</i> - no mitochondria so oxygen is not used within cell (and can be transported) ; - flexible / can squeeze through / can deform / can change shape, to pass through capillary / AW ; <i>context of biconcave or property of cell surface membrane</i> I 'squeeze between' unless elsewhere there is ref. to rbc within a capillary - biconcave ; (biconcave, so compared to spherical) increased, surface area / SA:V, for uptake of oxygen / gas exchange / AW ; (biconcave, so) reduce distance for diffusion of oxygen to, haemoglobin / hb ;	4